PVP 14

Code: CE2T1, ME2T1, CS2T1, IT2T1, EE2T1, EC2T1, AE2T1

I B.Tech - II Semester – Regular Examinations – JULY 2015

ENGINEERING MATHEMATICS - II
(Common for all Branches)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11x 2 = 22 M

1. a) Define Echelon form of a Matrix.
   b) Explain Normal form of a Matrix.
   c) Prove that If λ is an Eigen value of an orthogonal matrix then $\frac{1}{\lambda}$ is also its Eigen value.
   d) Find the Sum and product of the Eigen values of
   $$A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}.$$
   e) Find $L(\cos^2 2t)$
   f) Define Dirac's delta function.
   g) Find $L^{-1}\left(\cot^{-1}\left(\frac{s}{2}\right)\right)$.
   h) Write the existence conditions of Fourier series expansion.
   i) Define change of scale property of Fourier transform.
   j) State Damping rule of Z- transform.
   k) State Initial value theorem of Z-transform.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks. 16 x 3 = 48 M

2. a) Are the following vectors linearly dependent? If so find the relation between them, where X=(1,3,4,2), Y=(3,-5,2,2), and Z=(2,-1,3,2). 8 M

b) Investigate the values of λ and μ so that the equations $x+y+z=6$, $x+2y+3z=10$, $x+2y+\lambda z=\mu$ have 8 M
i) No solution ii) A unique solution
iii) An infinite number of solutions.

3. a) State the Cayley-Hamilton theorem for a matrix and find verify the Cayley-Hamilton for the given matrix also find A^{-1} where $A=\begin{bmatrix} 7 & 2 & -2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{bmatrix}$ 8 M

b) If $A=\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ then obtain the matrix P such that $P^{-1}AP$ is a diagonal matrix. 8 M

4. a) Evaluate $L\left[\int_0^t \frac{\cos at - \cos bt}{t} dt\right]$ 8 M

b) Solve the differential equation $y^{11} - 3y^{1} + 2y = e^{3t}$ where $y(0) = 1, y^{1}(0) = 0$ using transform method. 8 M

5. a) Obtain Fourier series for the function f(x) given by

$$f(x) = 1 + \frac{2x}{\pi}: \qquad -\pi \le x \le 0$$

$$1 - \frac{2x}{\pi} \qquad 0 \le x \le \pi$$

Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots = \frac{\pi^2}{8}$. 8 M

b) Find the Fourier Cosine transform of e^{-x^2}. 8 M

6. a) Find the Z-transform of

i) $3n - 4\sin\frac{n\pi}{4} + 5a$ ii) $\sin(3n + 5)$. 8 M

b) Using Z - transforms

solve $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ with $y_0 = y_1 = 0$. 8 M

PVP 14

Code: CE2T1, ME2T1, CS2T1, IT2T1, EE2T1, EC2T1, AE2T1

I B.Tech - II Semester – Regular Examinations – April 2016

ENGINEERING MATHEMATICS - II
(Common for all Branches)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11x 2 = 22 M

1.

a) Find the rank of $\begin{bmatrix} 1 & 4 & 5 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

b) Show that the vectors X_1=(2, -2, 1), X_2= (1, 4, -1), and X_3=(4, 6, -3) are linearly independent.

c) If $\lambda_1, \lambda_2 \ldots\ldots \lambda_n$ are Eigen values of a Matrix A then prove that A^{-1} has the Eigen values of $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \ldots\ldots \frac{1}{\lambda_n}$

d) Find the Eigen values of the matrix $\begin{bmatrix} 6 & 8 \\ 8 & -6 \end{bmatrix}$

e) Find Laplace transformation of $sin^3 2t$

f) Find Laplace transformation of $\frac{sin2t}{t}$

g) Find the value of a_0 for the function $f(x)=x$ *in* $0 < x < 2$

h) Define Fourier sine and cosine transforms of the function $f(x)$

i) Find the Fourier sine transformation of $\frac{1}{x}$

j) Evaluate the inverse Z-transformation of

$$e^{1/z} + \frac{z(z-\cos\theta)}{z^2+2z\cos\theta+1}$$

k) Evaluate $z\left[\frac{1}{(n+1)!}\right]$

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

16 x 3 = 48 M

2.

a) Solve the following equations by Gauss-Jordon method $x+2y+z-w=-2$, $2x+3y-z+2w=7$, $x+y+3z-2w=-6$, $x+y+z+w=2$. 8 M

b) Apply Gouss-Sidel iteration method to solve the equations. $20x+y-2z=17$, $3x+20y-z=-18$, $2x-3y+20z=25$ 8 M

3.

a) Find the characteristic equation of the matrix $A=\begin{bmatrix}2 & 1 & 1\\ 0 & 1 & 0\\ 1 & 1 & 2\end{bmatrix}$ and hence, find the matrix represented by $A^8-5A^7+7A^6-3A^5+A^4-5A^3+8A^2-2A+I$. 8 M

b) Find the Eigen values and Eigen vectors of the matrix $\begin{bmatrix}-2 & 2 & -3\\ 2 & 1 & -6\\ -1 & -2 & 0\end{bmatrix}$ 8 M

4.

a) Find the inverse Laplace Transform of $\frac{s}{s^4+s^2+1}$ 8 M

b) Solve the equation $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = e^{-t}\sin t$, $x(0) = 0$, $x'(0) = 1$ 8 M

5.

a) Obtain Fourier series for the function

$$f(x) = \begin{cases} \pi x & 0 \le x \le 1 \\ \pi(2-x) & 1 \le x \le 2 \end{cases}$$ 8 M

b) Find the Fourier transformation of

$$f(x) = \begin{cases} 1 - x^2 & if\ |x|<1 \\ 0 & if\ |x|>1 \end{cases}$$ and hence evaluate

$$\int_0^\infty \left(\frac{x\cos x - \sin x}{x^3}\right) \cos\frac{x}{2}\, dx$$ 8 M

6.

a) Use convolution theorem to evaluate $Z^{-1}\left[\frac{z^2}{(z-a)(z-b)}\right]$ 8 M

b) Solve $u_{n+2} + u_n = \cos\frac{n}{2}$ 8 M

PVP 14

Code: CE2T1, ME2T1, CS2T1, IT2T1, EE2T1, EC2T1, AE2T1

I B.Tech - II Semester – Regular/Supplementary Examinations
April - 2018

ENGINEERING MATHEMATICS - II

(Common for all Branches)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11 x 2 = 22 M

1.

a) Define Normal form of a matrix.

b) Write the working rule for finding the inverse of a matrix by using elementary row operations.

c) If eigen values of A are - 1, 2, -3 then find the eigen values of adj A.

d) Prove that eigen values of A and A^T are same.

e) Write the existence conditions of the Laplace Transforms of a function.

f) Find Inverse Laplace Transforms of $\frac{4}{(s+1)(s+2)}$

g) State and prove First shifting theorem of Laplace Transforms.

h) Find the finite Fourier sine transform of f(x) = x where 0<x<4

i) Define Periodic function with example.

j) Find $Z\left\{e^{t}\sin 2t\right\}$

k) State and Prove damping rule for z- transform.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

3 x 16 = 48 M

2. a) Solve the following system of equations

$$3x + y + z = 8,\ x + 2y - z = 5,\ 2x + y + 3z = 4$$

Using Gauss elimination method. 8 M

b) Show that the only real number λ for which the system

$$x + 2y + 3z = \lambda x,\ 3x + y + 2z = \lambda y,\ 2x + 3y + z = \lambda z$$

has non – zero solution is 6 and solve them when λ = 6.

8 M

3. a) Diagonalize the matrix A= $\begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$ 8 M

b) If A and B are n rowed square matrices and if A is invertable, show that $A^{-1}B$ *and* BA^{-1} have the same eigen values. 8 M

4. a) Find $L\left\{\int_0^t t\,e^{-t}\sin 4t\,dt\right\}$ 8 M

b) Using Laplace Transforms solve

$$\frac{d^2y}{dt^2} + 2.\frac{dy}{dt} + 5y = e^{-t}.\sin t$$

given that $y(0) = 0,\ y'(0) = 1$. 8 M

5. a) Find the Fourier series for $f(x) = x - x^2\ in\ [-\pi, \pi]$. 8 M

b) Find the Fourier cosine and sine transforms of $\frac{e^{-ax}}{x}$ and hence deduce that

$$\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} \sin sx\ dx = \tan^{-}\left(\frac{s}{a}\right) - \tan^{-}\left(\frac{s}{b}\right)$$ 8 M

6. a) Find $z\left\{(\cos\theta + i\sin\theta)^n\right\}$. Hence evaluate $z(\cos n\theta)\ and\ z(\sin n\theta)$ 8 M

b) Find $z^{-1}\left\{\frac{3z^2}{(5z-1)(5z+2)}\right\}$ 8 M

PVP 14

Code: CE2T1, ME2T1, CS2T1, IT2T1, EE2T1, EC2T1, AE2T1

I B.Tech - II Semester – Regular/Supplementary Examinations
April - 2019

ENGINEERING MATHEMATICS - II
(Common for all Branches)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11 x 2 = 22 M

1.

a) Define the following terms: i) Consistency ii) Trivial Solution for Homogeneous System of Equations.

b) For a non-homogeneous system, when we say the system is consistent and in what case we get infinite number of solutions?

c) Find the eigen values of $A^2 - 2A + I$,where

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 4 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

d) Two eigen values of the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ are equal to 1 each. Find the eigen values of A^{-1}.

e) Write the statement of Cayley-Hamilton theorem.

f) Find the Laplace transform of sin2tcos3t.

g) Find the inverse Laplace transform of $\frac{1}{s^2-5s+6}$.

h) Write the Fourier series for the function f(x) in the interval $-c < x < c$.

i) Define the Fourier transform and inverse Fourier transform of a function.

j) Write the convolution theorem in Z-transforms.

k) Show that $\boldsymbol{z}\left(\frac{1}{n+1}\right) = z \log\left(\frac{z}{z-1}\right)$.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

3 x 16 = 48 M

2. a) For what values of μ, λ the simultaneous equations

$x+y+z=6$

$x+2y+3z=10$

$x+2y+\lambda z=\mu$ have

i) No solution

ii) Unique solution

iii) Infinite number of solutions 8 M

b) Apply Gauss elimination method to solve the following system of equations

$x+4y-z=-5, \quad x+y-6z=-12, \quad 3x-y-z=4$ 8 M

3. a) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ and find its inverse. Also express

$A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ as a linear polynomial in A. 8 M

b) Find the eigen values and eigen vectors of the following matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$ 8 M

4. a) Find the Laplace transform of the following functions

i) $\frac{(e^{-at} - e^{-bt})}{t}$ ii) $f(t) = \begin{cases} 4, & when\ 0 \le t \le 1 \\ 3, & when\ t > 1 \end{cases}$ 8 M

b) Use Laplace transform to solve 8 M

$$\frac{d^2x}{dt^2} - 2\frac{dx}{dt} + x = e^t \ with\ x = 2, \frac{dx}{dt} = -1\ at\ t = 0$$

5. a) Find the Fourier series to represent $f(x)=e^x$, $-\pi<x<\pi$ and hence derive a series for $\frac{\pi}{\sin h\pi}$ 8 M

b) Find the Fourier sine and cosine transform of

$f(x) = \begin{cases} sinx, if\ 0 < x < a \\ 0, if\ x \ge a \end{cases}$ 8 M

6. a) If $z[f(n)] = \frac{5z^2+3z+12}{(z-1)^4}$, find f(2) and f(3) 8 M

b) Solve 8 M

$$u_{n+2} + 4u_{n+1} + 3u_n = 3^n \ given\ that\ u_0 = 0, u_1 = 1$$